Contextualizing Harm: Simulating Subjectivity in Automated Toxicity Detection

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1 Introduction

Automated toxicity detection systems are widely used to moderate online platforms, yet research consistently shows that perceptions of toxicity are inherently subjective (Talat, 2016; Davidson et al., 2017). Sap et al. (2021) demonstrates that demographic factors (including race, gender, and political ideology) and social attitudes toward free speech, racism, and empathy systematically influence how individuals label language as toxic or offensive.

Despite this evidence, most machine learning models operate under the assumption of a single, objective definition of harm (Gordon et al., 2022), ignoring the reality that toxicity judgments vary across communities and contexts. This "one-size-fits-all" approach frequently leads to unintended biases, such as the disproportionate flagging of African American English (AAE) as toxic (Sap et al., 2019; Dixon et al., 2018).

This project investigates whether Large Language Models (LLMs) can simulate these human differences by adapting model behavior based on annotator-specific personas. Building on recent work exploring how LLMs reflect demographic opinions (Argyle et al., 2022; Santurkar et al., 2023), we aim to move beyond single-label predictions to evaluate how different persona constructions (demographic-only, belief-only, and combined) influence model alignment with human toxicity judgments across multiple dimensions, including general toxicity, personal offense, and racism, intending to build systems that transparently represent variation in toxicity judgments across communities.

2 Experimental Setup

We utilize the human-annotated dataset from Sap et al. (2021), which provides granular annotator profiles, including demographics (e.g., age,

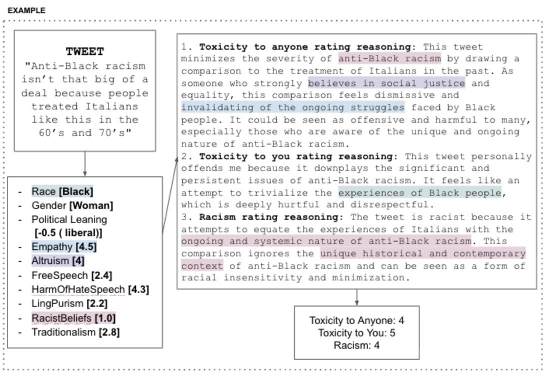

Figure 1: Example of Inputs and Output

gender, occupation), specific belief dimensions such as Racist Beliefs, Traditionalism, and Language Purism, alongside toxicity annotations of tweets. For example, an annotator may be described as a 24-year-old Black woman with high empathy and low endorsement of racist beliefs. To simulate these diverse perspectives, we employ `Qwen2.5-32B-Instruct` (Dubey et al., 2024).

Our experimental setup is as follows:

Persona Construction Prompting the model with "demographic-only", "belief-only", and "combined" (demographics + beliefs) profiles.

Evaluation Dimensions Models rate tweets on General Toxicity, Personal Offense, and Racism.

Prompting Strategies We compare Direct Questions (zero-shot prompting) against Chain-of-Thought (CoT) reasoning (Wei et al., 2022) to assess whether explanations improve alignment with human annotators.

3 Results

We evaluated model performance using F_1 scores against human annotations across three evaluation dimensions: Toxicity to Anyone, Toxicity to You, and Racism. Our experiments reveal that different persona constructions excel at different toxicity detection tasks, suggesting that the type of contex-

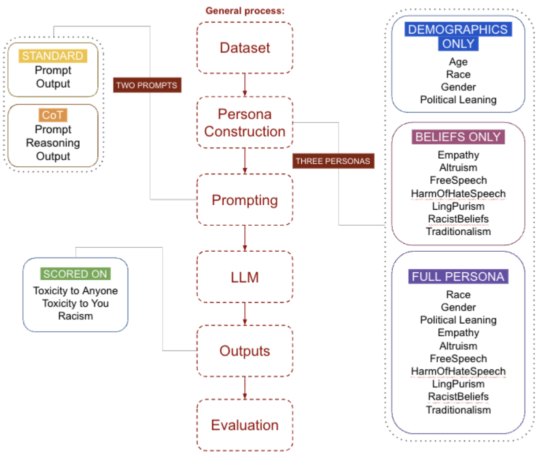

Figure 2: General Simulation Process

tual information provided to the model significantly affects alignment with human judgments.

As shown in Figures 3 and 4, performance across persona constructions varied depending on both the evaluation task and prompting strategy. Under standard prompting, belief-only personas achieved the strongest alignment on general toxicity detection tasks, while demographic-only personas performed best on personal offense judgments. Full personas, which combine demographic and belief-based attributes, consistently achieved the strongest performance on racism detection tasks.

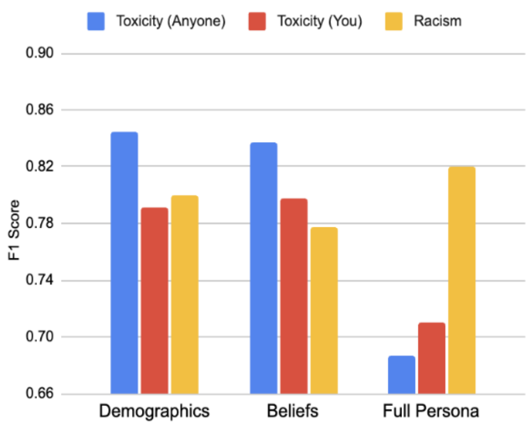

Figure 3: F_1 scores across demographic-only, belief-only, and full personas under standard prompting. Different persona constructions achieve stronger alignment across different toxicity evaluation tasks.

Under Chain-of-Thought (CoT) prompting, several of these trends shifted. Demographic-only personas improved on general toxicity detection tasks, while belief-only personas performed comparably to or slightly better than demographic personas on personal offense judgments. In contrast to these shifts, full personas remained the strongest performers on racism detection tasks across both prompting strategies. In particular, full personas combined with Chain-of-Thought prompting achieved the highest racism detection performance overall ($F_1 \approx 0.889$).

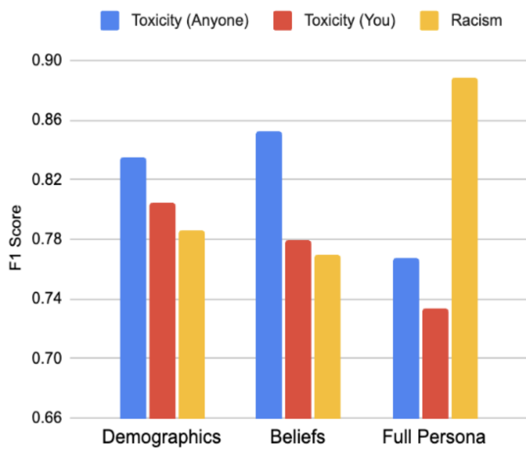

Figure 4: F_1 scores across demographic-only, belief-only, and full personas under Chain-of-Thought prompting. Full personas achieve the strongest performance on racism detection tasks.

To better understand how reasoning-based prompting affects different persona constructions, we directly compared standard and Chain-of-Thought prompting within each persona type. As shown in Figures 5, 6, and 7 the effects of Chain-of-Thought prompting were highly persona-dependent.

In contrast, full personas exhibited substantially larger gains under Chain-of-Thought prompting, particularly for racism detection tasks. This suggests that richer persona representations may benefit more from reasoning-based prompting than simpler demographic-only or belief-only personas.

These findings suggest that the effectiveness of Chain-of-Thought prompting depends heavily on both persona complexity and the type of toxicity judgment being modeled. However, the strong and consistent performance of full personas on racism detection across both prompting strategies indicates that perceptions of racist language may rely more heavily on the interaction between demographic identity and social beliefs than broader toxicity judgments.

However, increasing the amount of persona information did not consistently improve alignment with human annotations. While full personas improved performance on identity-targeted hate and racism-

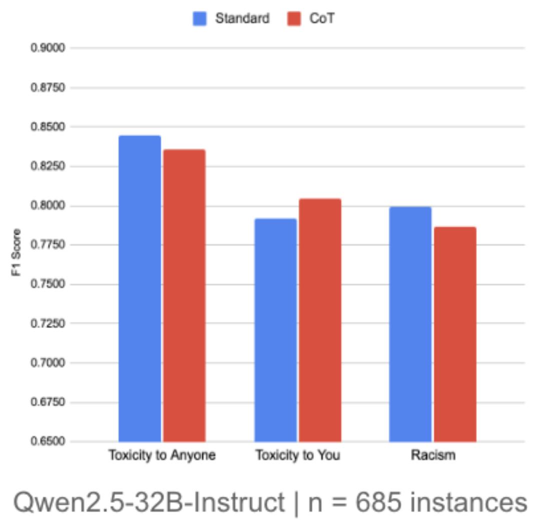

Figure 5: Comparison of standard and Chain-of-Thought prompting for demographic-only personas. CoT prompting produces only minor performance changes across toxicity evaluation tasks.

Figure 6: Comparison of standard and Chain-of-Thought prompting for belief-only personas. CoT prompting slightly improves general toxicity detection while reducing performance on other tasks.

related tasks, they often reduced performance on broader toxicity evaluation tasks. This suggests that richer persona representations may introduce redundant or conflicting signals that reduce model alignment with human annotations.

Additionally, demographic attributes and belief dimensions are not fully independent. Prior work by Sap et al. (2021) found correlations between demographic identity and attitudes such as traditionalism, endorsement of racist beliefs, empathy, and views on free speech. These relationships may help

Figure 7: Comparison of standard and Chain-of-Thought prompting for full personas. CoT prompting substantially improves racism detection and general toxicity alignment for combined personas.

explain why combining demographic and belief information into a single persona did not universally outperform simpler persona constructions.

To further investigate this relationship, we prompted the LLM to predict annotator belief scores using demographic information alone and evaluated the alignment between predicted and ground-truth belief dimensions. As shown in Figure 8, several belief attributes — particularly empathy, altruism, and perceived harm of hate speech — could be inferred with relatively high accuracy from demographic personas alone.

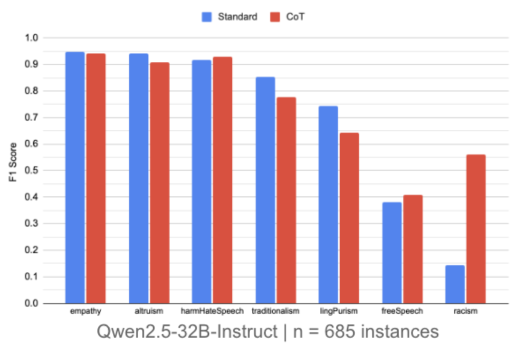

Figure 8: F_1 scores for predicting belief dimensions from demographic personas under standard and Chain-of-Thought prompting. Strong performance across several belief categories suggests that demographic information partially encodes underlying social attitudes.

These findings mirror patterns observed in Sap et al. (2021), where both demographic identity and social beliefs jointly influenced toxicity judgments. For example, conservative annotators and annotators scoring highly on *FREEOFFSPEECH* and *TRADITIONALISM* were more likely to rate African American English (AAE) posts as offensive or racist. Conversely, annotators with stronger *HARMOFHATESPEECH*, empathy, and altruism scores were more likely to identify anti-Black content as harmful or toxic. Similar trends were also observed for anti-Black and vulgar language categories, where political ideology and belief-based attributes significantly affected perceived offensiveness and racism ratings.

Importantly, these relationships may help explain why full personas achieved the strongest performance on racism detection tasks. Because perceptions of racist language appear to depend on both demographic identity and belief-based attributes simultaneously, combining these signals into a unified persona may provide a more complete representation of the factors underlying human racism annotations. This contrasts with broader toxicity evaluation tasks, where additional persona information often introduced redundant or conflicting signals that reduced model alignment with human annotations.

Taken together, our results suggest that persona-conditioned LLMs may partially reproduce the same demographic-belief relationships observed in human annotators. This further supports the broader claim that toxicity judgments are socially and ideologically situated rather than governed by a single universal definition of harm.

4 Discussion

Our results suggest that different forms of toxicity rely on different underlying social signals and therefore benefit from different types of contextual modeling. In particular, racism detection consistently benefited from combining demographic and belief-based persona information, while broader toxicity evaluation tasks often performed better with simpler persona representations. These findings indicate that identity-targeted harms may require richer social and ideological context than generalized toxicity classification tasks.

In particular, full personas substantially improved racism detection performance, especially under Chain-of-Thought prompting ($F_1 \approx 0.889$ compared to approximately 0.77–0.79 for partial personas). This reinforces the idea that perceptions of identity-targeted language depend heavily on the interaction between demographic identity and social beliefs. Prior work by Sap et al. (2021) similarly found that both demographic factors and belief-based attributes jointly influenced racism annotations, particularly for African American English (AAE) and anti-Black language. Our results suggest that persona-conditioned LLMs may partially reproduce these same demographic-belief relationships when simulating annotator judgments.

At the same time, richer persona information did not universally improve performance. While full personas improved racism detection, they often reduced alignment on broader toxicity evaluation tasks, implying that additional persona information can introduce redundant or conflicting signals. Our belief-prediction experiments further support this interpretation by showing that several social belief dimensions can already be inferred from demographic information alone. This further supports the idea that richer persona representations are most useful for modeling identity-targeted harms rather than broader toxicity judgments.

More broadly, these findings have implications for the design of future moderation and toxicity detection systems. Current toxicity classifiers often assume a single universal definition of harm, despite substantial variation in how different communities interpret offensive language. Our results suggest that systems designed to evaluate identity-targeted harms may benefit from incorporating both demographic and belief-based contextual information, while broader toxicity classification tasks may require simpler or more targeted persona representations. Rather than treating all forms of toxicity identically, future moderation systems may need to adapt to the specific social dimensions most relevant to the type of harm being evaluated.

5 Limitations and Future Work

While our experiments demonstrate promising results, several limitations remain. First, our study relies primarily on a single dataset derived from Sap et al. (2021), which focuses on specific demographic and ideological dimensions within English-language online discourse. As a result, our findings may not generalize to other cultural contexts, languages, or forms of identity-targeted harm. Additionally, persona representations in our experiments are simplified abstractions of human identity and belief systems. Real-world perceptions of toxicity are shaped by significantly more complex social, cultural, and contextual factors than can be captured through prompt-based personas alone.

Another limitation is that persona-conditioned LLM outputs may reflect stereotypes or biases present in the model's pretraining data rather than genuine perspective-taking capabilities. High alignment with human annotations does not necessarily imply that the model truly understands social context or harm. Furthermore, our experiments primarily focus on a single model family and limited prompting strategies, which constrains our ability to draw broader conclusions about persona simulation across architectures and scales.

Future work should explore richer persona representations, larger and more diverse datasets, and broader evaluations across multiple model families. Expanding beyond demographic and belief-based personas to include contextual, cultural, or community-specific factors may further improve the modeling of subjective toxicity judgments. Additionally, future systems could move beyond single-label toxicity classification toward distributional or multi-perspective outputs that explicitly represent disagreement and variation across annotator groups. Such approaches may lead to moderation systems that are more transparent, equitable, and socially aware.

6 Conclusion

In this work, we explored whether LLMs can simulate subjective toxicity judgments through persona-conditioned prompting. Our results show that different persona constructions excel at different toxicity evaluation tasks: demographic-only personas aligned more closely with personal offense judgments, belief-only personas performed best on broader toxicity detection tasks, and full personas consistently achieved the strongest racism detection performance. These findings suggest that the effectiveness of persona conditioning depends not only on the amount of contextual information provided, but also on the specific type of toxicity being modeled.

We further found that combining demographic and belief-based information was particularly beneficial for modeling identity-targeted harms, while richer persona representations often introduced conflicting or redundant signals for broader toxicity classification tasks. Additionally, our experiments suggest that demographic information can partially encode underlying social beliefs, helping explain the interaction between persona complexity and model performance.

Overall, our findings highlight the limitations of treating toxicity detection as a single universal classification task. Instead, different forms of harmful language may require different types of contextual and social information to model human judgments effectively. As LLM-based moderation systems continue to develop, incorporating multiple perspectives and forms of disagreement may be essential for building more transparent and socially aware toxicity detection systems.

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